

STA 35C: Statistical Data Science III

Lecture 24: Clustering & K-means Clustering

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Spring 2026, UC Davis

Announcement

Final exam: Fri, June 5 (1:00 pm–3:00 pm) in Wellman Hall 6 (=classroom)

- **Be on time:** The exam starts at 1:00 pm and ends at 3:00 pm sharp
- **Three hand-written cheat sheets allowed:** Letter-size (8.5"×11"), double-sided, brief formulas/notes
- **Calculator:** A simple (non-graphing) scientific calculator is allowed
- **No other materials** beyond the cheat sheets (no textbooks, etc.)
- **SDC accommodations:** Confirm scheduling with AES online as soon as possible

Course evaluation: Please share your feedback comments by Thu, June 4

Agenda

Quick recap: Principal component analysis (PCA)

- Objective: dimension reduction with minimal information loss
- Intuition: projection that retains maximum variance
- Proportion of variance explained & choosing number of PCs via scree plot

Today: Clustering

- Clustering problem
- Overview of two methods: K-means clustering & hierarchical clustering
- K-means clustering
 - Intuition & problem formulation
 - Algorithm
 - Illustration
 - Assessment

Recap: PCA

Problem setup:

- We have data of $X \in \mathbb{R}^p$, with potentially large dimension p
- **Goal:** reduce dimension from p to $r \ll p$ while retaining most of the “information”

PCA approach:

- Project data (X) onto an r -dimensional subspace (spanned by r vectors)
- These r principal components are chosen to capture maximum variance in X
 - **First PC:** a unit vector $\mathbf{u}_1 \in \mathbb{R}^p$ maximizing variance:

$$\mathbf{u}_1 = \operatorname{argmax}_{\|\mathbf{u}\|=1} \frac{1}{n} \sum_{i=1}^n (\mathbf{u} \cdot \mathbf{x}_i)^2$$

- Subsequent PCs $\mathbf{u}_2, \dots, \mathbf{u}_p$ are found similarly, each orthogonal to all previous PCs

Result:

- Often the first few PCs ($r \ll p$) capture most of the variation
- This allows dimension reduction by using only $(Z_{i1}, \dots, Z_{ir})$ for observation i

Recap: PC scores, PVE, and choosing number of PCs

PC scores: PCA is a change of basis (=change of coordinate system)

- The k -th **PC score** of $\mathbf{x}_i$ is

$$Z_{ik} = \langle \mathbf{u}_k, \mathbf{x}_i \rangle = \sum_{j=1}^p u_{kj} x_{ij}.$$

- These Z_{ik} values become the coordinates of $\mathbf{x}_i$ in the new (PC) coordinate system

Proportion of variance explained (PVE):

- Total variance: $\text{Var}(X) = \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i\|^2 = \sum_{j=1}^p \text{Var}(X_j) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^p x_{ij}^2$
- Variance explained by the k -th PC: $\text{Var}(\langle \mathbf{u}_k, X \rangle) = \frac{1}{n} \sum_{i=1}^n Z_{ik}^2$
- $\text{PVE}_k = \frac{\text{Var}(\mathbf{u}_k \cdot X)}{\text{Var}(X)}$ and $\text{PVE}_{1:r} = \sum_{k=1}^r \text{PVE}_k$

Choosing r : Use a scree plot or the cumulative PVE to decide how many PCs to keep

Clustering: Problem setup

Data:

$$\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}, \quad \mathbf{x}_i \in \mathbb{R}^p.$$

Goal: Partition observations into clusters so that

- observations in the same cluster are similar,
- observations in different clusters are dissimilar.

Examples where clustering may be useful:

- **Cancer subtyping:** group patients or tumor samples using gene-expression profiles
- **Market segmentation:** group customers by purchasing patterns or behavior
- **Document analysis:** group articles by word usage
- **Image analysis:** group pixels or image patches by color/texture features

Important distinction from classification:

- Classification: labels/classes are observed in training data.
- Clustering: labels/classes are unknown; the algorithm discovers possible groups.

Illustration of clustering problem

Setup:

- Data: $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subset \mathbb{R}^p$
- **Goal:** Partition the observations into clusters so that points within each cluster are “similar,” while points in different clusters are “different”

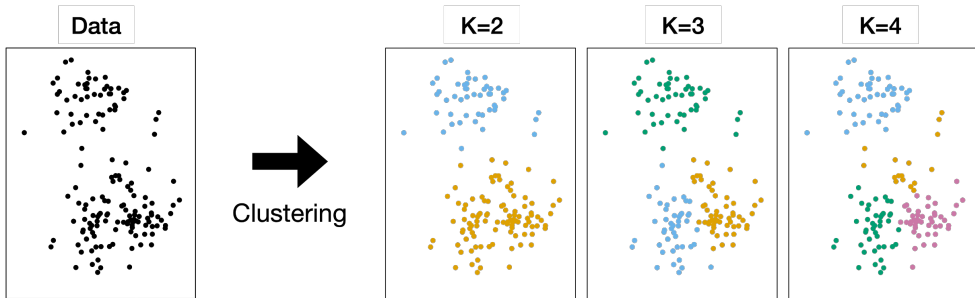


Figure: Illustration of clustering. Given a dataset of X (**Left**), we want to partition the observations into K distinct clusters (**Right**).

Distance and standardization matter

Clustering requires a notion of similarity. A common choice is Euclidean distance:

$$d(\mathbf{x}_i, \mathbf{x}_{i'}) = \|\mathbf{x}_i - \mathbf{x}_{i'}\|_2.$$

Scaling issue:

- Variables with larger numerical scales can dominate distances.
- Example: income in dollars may dominate age in years.

Common practice:

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}.$$

- Standardize before clustering when variables have different units or scales.
- The choice of distance and scaling can substantially affect the clustering result.

Clustering: Overview of two algorithms

There are many clustering methods, but we focus on two well-known approaches:

- **K-means clustering** (Today)
 - Choose K in advance, then partition observations into K clusters
 - Simple & well-suited for relatively spherical clusters in a feature space
- **Hierarchical clustering** (next lecture)
 - We do not specify the number of clusters upfront
 - Observations are successively merged or split to form a hierarchical tree structure (*dendrogram*)
 - We can then *cut* the tree at various levels to obtain different numbers of clusters

K-means clustering: Basic idea

Goal: Partition the data $\{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subset \mathbb{R}^p$ into K non-overlapping clusters

- We specify the desired number of clusters K in advance
- Partition indices $\{1, \dots, n\}$ into $C_1, \dots, C_K$ with:
 - $C_1 \cup C_2 \cup \dots \cup C_K = \{1, \dots, n\}$: every $\mathbf{x}_i$ belongs to at least one of the K clusters
 - $C_k \cap C_{k'} = \emptyset$ for all $k \neq k'$; each $\mathbf{x}_i$ belongs to at most one cluster

Formulation of K-means clustering problem

$$\text{minimize}_{C_1, \dots, C_K} \left\{ \sum_{k=1}^K W(C_k) \right\}$$

- The quantity $W(C_k)$ measures within-cluster variation
 - We want the clusters to be “tight,” so $\sum_{k=1}^K W(C_k)$ to be as small as possible
- K-means clustering typically uses the squared (Euclidean) distance

$$W(C_k) = \frac{1}{|C_k|} \sum_{i, i' \in C_k} \|\mathbf{x}_i - \mathbf{x}_{i'}\|^2 = \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2$$

Example: Within-cluster variation

Example

Let $\mathcal{X} = \{\mathbf{x}_1 = (-2, 1), \mathbf{x}_2 = (-1, 3), \mathbf{x}_3 = (2, 0), \mathbf{x}_4 = (3, -2)\} \subset \mathbb{R}^2$. Let $K = 2$ and

$$C_1 = \{1, 2\}, \quad C_2 = \{3, 4\}.$$

Recall

$$W(C_k) = \frac{1}{|C_k|} \sum_{i, i' \in C_k} \|\mathbf{x}_i - \mathbf{x}_{i'}\|^2.$$

For each cluster, the within-cluster variation can be computed as

$$W(C_1) = \frac{1}{2} [\|(-2, 1) - (-1, 3)\|^2 + \|(-1, 3) - (-2, 1)\|^2] = \frac{1}{2} \times (5 + 5) = 5,$$

$$W(C_2) = \frac{1}{2} [\|(2, 0) - (3, -2)\|^2 + \|(3, -2) - (2, 0)\|^2] = \frac{1}{2} \times (5 + 5) = 5.$$

Therefore, the K-means clustering objective value is $W(C_1) + W(C_2) = 5 + 5 = 10$.

Pop-up quiz: K -means objective

Using the four points

$$\mathbf{x}_1 = (-2, 1), \quad \mathbf{x}_2 = (-1, 3), \quad \mathbf{x}_3 = (2, 0), \quad \mathbf{x}_4 = (3, -2),$$

compare two possible $K = 2$ clusterings:

$$\text{A: } C_1 = \{1, 2\}, \quad C_2 = \{3, 4\}, \quad \text{B: } C_1 = \{1, 4\}, \quad C_2 = \{2, 3\}.$$

Question: Which clustering would the K -means objective prefer?

- A) Clustering A, because points within each cluster are closer to their centroid.
- B) Clustering B, because each cluster contains one left point and one right point.
- C) Both are equally good because both have two clusters of size two.
- D) Cannot be determined because no response variable Y is available.

Answer: A. The K -means objective prefers smaller within-cluster variation. Clustering A groups nearby points and has much smaller within-cluster variation than clustering B.

K-means clustering: Algorithm

K-means clustering is a hard combinatorial problem, so we use a heuristic algorithm:

K-means clustering algorithm

1 **Initialize:** Randomly assign each of the n observations to one of K clusters

2 **Iterate until assignments stop changing:**

(a) **Update the centroids.** For each cluster C_k , compute the centroid

$$\bar{\mathbf{x}}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i.$$

(b) **Reassign.** For each observation i , reassign it to the cluster whose centroid is closest in squared Euclidean distance

- Each iteration reduces the objective but may converge to a local optimum
- Often repeated from multiple random starts to choose the best result

Example: One iteration of K-means clustering

Example

Data: $\mathcal{X} = \{(-2, 1), (-1, 3), (2, 0), (3, -2)\}$, $K = 2$. Suppose $K = 2$, and the the *initial* cluster assignment is

$$C_1 = \{1, 4\}, \quad C_2 = \{2, 3\}.$$

Step (a): Compute centroids.

$$\bar{\mathbf{x}}_1 = \frac{1}{2}[(-2, 1) + (3, -2)] = (0.5, -0.5), \quad \bar{\mathbf{x}}_2 = \frac{1}{2}[(-1, 3) + (2, 0)] = (0.5, 1.5).$$

Step (b): Reassign each point to the closer centroid.

- $\mathbf{x}_1$ is closer to $\bar{\mathbf{x}}_2$ because

$$\|\mathbf{x}_1 - \bar{\mathbf{x}}_1\|^2 = (-2.5)^2 + (1.5)^2 = 8.5 > \|\mathbf{x}_1 - \bar{\mathbf{x}}_2\|^2 = (-2.5)^2 + (-0.5)^2 = 6.5.$$

- Similarly, we observe $\mathbf{x}_2$ is closer to $\bar{\mathbf{x}}_2$, whereas $\mathbf{x}_3, \mathbf{x}_4$ are closer to $\bar{\mathbf{x}}_1$.

We get $C_1 = \{3, 4\}$, $C_2 = \{1, 2\}$.

Repeat **(a)** and **(b)** until the algorithm converges.

K-means clustering: Illustration of the algorithm iterations

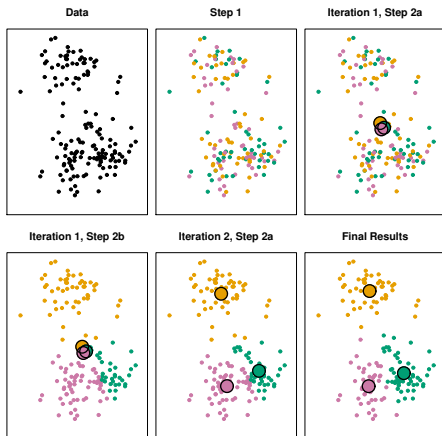


Figure: An example of the K-means with $K = 3$ over two iterations. Each iteration updates centroids (colored disks) and reassigns points [JWHT21, Figure 12.8].

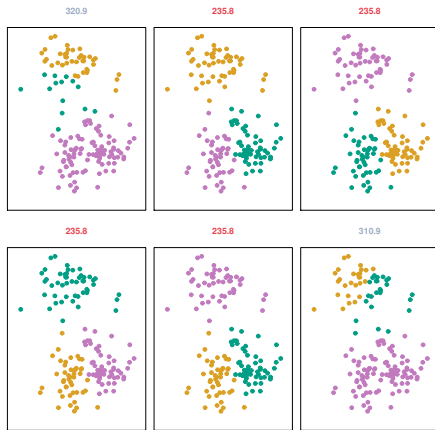
Steps:

- 1 **Initialize** random cluster labels
- 2 **Iterate:**
 - (a) Update cluster centroids
 - (b) Reassign points to nearest centroid

Remarks:

- Each iteration reduces the objective
- Final solution depends on initialization

K-means clustering: Local optima and multiple runs



Key points:

- K-means can converge to a suboptimal (local) solution
- Different initial cluster assignments can yield different final partitions
- Usually, re-run with multiple random starts and pick the best (lowest objective)

Figure: K-means with $K = 3$ repeated six times on the same data, each with a different random initial assignment. Above each plot is the final objective. Multiple local optima are found; the best has objective=235.8 [JWHT21, Figure 12.9].

Choosing the number of clusters K

Challenge: K -means requires K to be specified in advance.

Important fact:

The best achievable within-cluster variation cannot increase as K increases.

- If $K = n$, each point is its own cluster and within-cluster variation is 0.
- Therefore, the K -means objective alone cannot justify choosing the largest K .

Common approaches:

- **Elbow plot:** look for a point where increasing K gives diminishing returns.
- **Domain knowledge:** choose K based on interpretability or scientific context.
- **Stability:** check whether clusters persist under resampling or perturbations.

K-means clustering: Strengths and limitations

Strengths:

- Simple and computationally fast, especially for large data
- Often yields sensible clusterings if K is well-chosen
- Easy to interpret: each cluster has a centroid

Limitations:

- Must pre-specify the number of clusters K
- Can converge to a *local* rather than global optimum
- Assumes clusters are roughly spherical around centroids
- Sensitive to feature scaling, outliers, initialization, and non-spherical cluster shapes

Pop-up quiz: Choosing K

Suppose we run K -means on the same dataset for $K = 1, 2, 3, \dots$, always using the best result among many random starts.

Question: Which statement is most accurate?

- A) The within-cluster variation must increase as K increases.
- B) The within-cluster variation cannot increase as K increases, but this does not mean the largest K is the best choice.
- C) The best K is always $K = n$, because the objective becomes zero.
- D) K -means chooses K automatically, so no decision is needed.

Answer: B. Increasing K gives the algorithm more flexibility, so the within-cluster objective cannot increase. But $K = n$ is not useful for finding meaningful structure, so K must be chosen using judgment, elbow plots, stability, or domain knowledge.

Wrap-up: Takeaways

Clustering problem:

- We have feature vectors $\mathbf{x}_i \in \mathbb{R}^p$ (no response Y)
- **Goal:** partition observations into “clusters” so that points in the same cluster are similar, and points in different clusters are dissimilar

K-means clustering:

- Fix the number of clusters K in advance
- Define non-overlapping clusters $C_1, \dots, C_K$ to *minimize* the total within-cluster variation
- **Algorithm:**
 - i) *Initialize* random cluster assignments
 - ii) Iteratively (a) *update centroids*, and (b) *reassign points* until convergence
- **Limitations:** can get stuck in local optima; requires K pre-specified

Next time:

- Hierarchical clustering (no need to specify K)
- Dendrograms and various linkage criteria

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.